The Homology Cliff in Frozen ESM-2 Biosecurity Retrieval: Characterization, Null Controls, Mechanism, and a Deployable Rescue

Santiago Maniches*

April 12, 2026

Abstract

Frozen ESM-2 cosine k-NN retrieval on a 24,885-protein biosecurity-relevant test set exhibits a close-distant F_1 cliff that grows monotonically with both model scale and reference panel size. At scale t30 (150M parameters), R=1000, k=25, close-stratum $F_1 = 0.866$ against distant-stratum $F_1 = 0.120$ (gap +0.745). We report six pre-registered experiments on this phenomenon. (1) A 3000-cell main factorial confirms the cliff, with all 300 groups surviving a seed-variance gate. (2) A panel-label-shuffle null control (3000 cells) fails the pre-registered "distant F_1 at chance" criterion, diagnosed as retention of the 28.7% test class prior. (3) A pre-registered addendum null (SHA256 f3864d09...), permuting the full 24,885 label vector, produces gap $\approx 0\pm$ seed variance in every one of 300 groups, confirming the cliff is not a stratification artifact. (4) The cliff is a *precision* collapse, not a recall collapse: only 3 of 24,885 test proteins are true-positive, distant-stratum, predicted-negative, and distant-stratum positive-prediction precision collapses to 0.068 (44 predictions, 3 correct). Expected Calibration Error rises 4× from close (0.069) to distant (0.294). (5) Neither Ledoit-Wolf Mahalanobis whitening (180 cells), Fisher-Rao within-class whitening (180 cells), nor a stratified cosine+Mahalanobis cascade (180 cells) rescues the cliff; each degrades pooled F_1. (6) A panel-only triplet-surrogate linear projection, fit in under 5 seconds with no test leakage, *does* rescue: it wins pooled F_1 in 18 of 18 factorial groups and delivers $F_1 = 0.891$ at t30 R=1000 k=25, the highest value observed anywhere, improving distant-stratum F_1 by 48% relative (0.120 → 0.177). All code, four pre-registrations, 6000 per-cell results, and 180-cell factorials released as reproducibility package.

1 Introduction and contributions

Frozen protein language model embeddings increasingly feed biosecurity retrieval pipelines. If the embedding captures functional mechanism rather than sequence similarity, retrieval succeeds on novel threats; if not, it fails precisely where it matters. We quantify this failure mode (the *homology cliff*) with factorial rigor and pre-registered null controls, diagnose the mechanism, and identify a deployable rescue.

2 Methods

2.1 Data and embeddings

24,885 proteins (7,133 positive, 17,752 negative; 28.7% positive prior) from `experiment2_proteins_25k_filtered.`
Frozen ESM-2: t6 (8M, 320d), t12 (35M, 480d), t30 (150M, 640d). All embeddings L_2-normalized.

*ORCID 0009-0005-6480-1987. TOPOLOGICA LLC.

2.2 Factorial

Pre-reg SHA256 139f6012.... Panel size $R \in \{50, 100, 250, 500, 1000\}$, class-balanced. Neighbors $k \in \{1, 3, 5, 10, 25\}$. Metrics: cosine, Euclidean, Mahalanobis (Ledoit-Wolf + eigendecomposition whitening), learned (panel-only 50-iteration Adam linear projection to $\min(128, D)$ dim, triplet-surrogate loss, seed 0). Ten seeds (20260410..20260419). Stratification: close $s_{\max} \geq 0.95$, moderate $[0.90, 0.95)$, distant < 0.90. Per-cell F_1 with 10,000-resample percentile bootstrap CI. Seed-variance gate voids groups where $\sigma^{(10)}_{\text{distant}} \geq$ close-distant gap.

2.3 Null controls

Two nulls, both pre-registered. (A) Panel-only shuffle: permute panel labels, test labels real. (B) Full-pool permutation (addendum pre-reg SHA256 f3864d09...): permute the 24,885-entry label vector once per seed using `default_rng(seed + 7,777,777)`. Both panel and test labels random; stratification by $s_{\max}$ unchanged (label-free).

3 Results: main factorial

The cliff is monotone in scale and panel size. At k=25 cosine: gap rises from +0.289 (t6 R=50) to +0.745 (t30 R=1000). All 300 main groups pass the seed-variance gate.

scale	R	k	metric	close F_1	distant F_1
t6	1000	25	cosine	0.882	0.297
t12	1000	25	cosine	0.883	0.173
t30	1000	25	cosine	0.866	0.120

3.1 Cosine equals Euclidean on L_2-normalized embeddings

Proposition 1. *For finite $X \subset S^{d-1}$ and $y \in S^{d-1}$, cosine-k-NN and Euclidean-k-NN return identical neighbor sets.*

Proof. $\|x - y\|^2 = 2 - 2\langle x, y\rangle$ on S^{d-1}; monotone in cosine. □

Verified to 4 decimals in 75 of 75 cells.

4 Results: null controls

4.1 Panel-shuffle: FAILED pre-reg criterion

Across 3000 `negctrl_*.npz` cells, shuffled-panel close-distant gaps range +0.20 to +0.44 with stratum F_1 values far from chance. Diagnosis: the implementation permutes only panel labels, preserving the 28.7% test class prior and stratum-specific class composition.

4.2 Full-pool permutation: PASSED pre-reg criterion in 300/300 groups

With both panel and test labels permuted, gap $\in [-0.027, +0.038]$ with $\sigma \in [0.006, 0.116]$ across all 300 groups. Every 95% CI includes zero. At t30 R=1000 k=25 cosine: close $F_1 = 0.015$, distant $F_1 = 0.025$, gap -0.010 ± 0.031. The cliff is not a stratification artifact.

5 Mechanism: precision collapse

At t30 R=1000 k=25 cosine (seed 20260410), of 24,885 test proteins, **only 3** are true-positive, distant-stratum, predicted-negative. Distant-stratum positive-prediction precision is $3/44 = 0.068$ (44 positive predictions, 3 correct). Expected Calibration Error rises from 0.069 (close) to 0.154 (moderate) to **0.294** (distant). In the vote-fraction bin $[0.9, 1.0)$ on distant, 3 predictions, 0 correct: the classifier is confidently wrong where confidence matters. The cliff is a precision collapse driven by distant-stratum false alarms, not a recall collapse.

6 Results: metric ablations

6.1 Mahalanobis reduces the GAP but not by rescuing distant

At t30 R=1000 k=25: cosine (0.866, 0.120) $\rightarrow$ Mahalanobis (**0.456**, **0.087**) per (close, distant). The "+0.376 gap reduction" is entirely close-collapse; distant goes *down*. Mahalanobis degrades both strata.

6.2 Fisher-Rao within-class whitening is worse than Mahalanobis

180 cells pre-registered. Fisher gap exceeds Mahalanobis in 17 of 18 groups by +0.015 to +0.285. Replicates DR-010 direction. Companion paper: `fisher_negative_result.pdf`.

6.3 Stratified cosine+Mahalanobis cascade fails

Pre-registered H1: cascade beats max(cosine, Mahalanobis, learned) pooled F_1 by +0.02. Rejected 18/18 groups; cascade loses by −0.046 to −0.310. Companion paper: `cascade_negative_result.pdf`.

6.4 Learned projection WINS every stratum at every scale

R=1000 k=25	close F_1	mod F_1	dist F_1	pooled F_1
cosine t30	0.866	0.393	0.120	0.848
mah t30	0.456	0.118	0.087	0.435
fisher t30	0.478	0.130	0.094	0.461
learned t30	**0.901**	**0.525**	**0.177**	**0.891**

Wins 18 of 18 factorial groups. Distant F_1 improvement: +16% (t6), +33% (t12), +**48**% (t30). Companion: `learned_projection_rescue.pdf`.

7 Deployment recommendation

For frozen-PLM biosecurity retrieval: (1) fit a panel-only learned projection (5 seconds CPU, no test leakage), (2) route all distant-stratum positive hits to human review regardless of metric (calibration is poor), (3) consider panel augmentation from Mapper-identified positive-enriched manifold regions (149 Mapper nodes; top-6 all-positive; see Appendix).

8 Limitations

(L1) t33 (650M) not embedded. (L2) Only 425 of 24,885 accessions have Pfam annotations in current release; cross-family analysis deferred. (L3) Threshold sensitivity sweep not run. (L4) Learned projection calibration not measured.

9 Reproducibility

Four pre-registrations with SHA256 hashes. 6000 main+negctrl+fullnull per-cell results (10,000-resample bootstrap CIs per stratum per cell). 540 ablation cells (180 Mahalanobis + 180 Fisher + 180 cascade). Master tar: `TOPOLOGICA_homology_cliff_session_20260412.tar.gz`.

References

[1] Z. Lin et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023.

[2] O. Ledoit and M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *J. Multivariate Anal.*, 88:365–411, 2004.